

Diode laser with optical feedback

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Abstract

In this report we investigate the effect of optical feedback when using a laser diode with three different setups; one without optical feedback, the Littrow configuration and the Littman configuration. We characterize the laser diode and extract the lasing threshold current for all three setups. As expected we notice a decrease in lasing threshold current with optical feedback of ≈ 1 mA. We further calculate the laser wavelength using angle measurements and the grating equation for both setups with optical feedback, leading us to $\lambda_{\text{Littrow}} = 704(79)$ nm and $\lambda_{\text{Littman}} = 672(12)$ nm. Both values are consistent with the spec value of 670 nm [1], with the Littman measurement being more accurate by a factor of 7. Lastly we investigate the tunability of the laser in the Littman configuration by measuring the projected spacing of different lasing modes and extracting the mean mode spacing $\overline{\Delta\nu} = 95(9)$ GHz, as well as the total frequency range 1330(130) GHz. Using these measurements we can estimate the free spectral range (FSR) of a Fabry-Perot interferometer by converting the time-domain measurements to frequencies, yielding an FSR of $\nu_{\text{FSR}} = 980(210)$ GHz. This is consistent with the possible FSR range given in [2].

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0 Introduction

Lasers are extremely useful tools in many fields of research and allow for advanced technical applications. They produce coherent light beams of a specific frequency, providing great possibilities in almost all fields of science, including physics, chemistry or medicine. Lasers, for example, revolutionized the field of spectroscopy, allowing us to intentionally probe atoms and molecules at certain frequencies to gather new insights into the inner workings.

However, the output light of a diode laser made of a semiconductor NP-junction is not completely coherent, as different modes at different frequencies are supported. At first sight, that sounds like a disadvantage of diode lasers in comparison to others like atomic lasers, but it offers the possibility to tune the wavelength.

In this experiment, the emitted mode of the laser is tuned with external feedback. This means that one certain selected mode of the emitted laser light is directed back into the diode, to amplify the intensity of this mode via stimulated emission.

In this experiment external feedback is achieved in two configurations, the Littman and the Littrow configuration. The output of the laser is then characterized by measuring the characteristic intensity-to-current curve of both configurations as well as the laser without feedback mechanism. Additionally, the range over which the wavelength can be tuned is estimated by switching through the optical modes in the Littman configuration. Lastly, also a Fabry-perot interferometer is used and its free spectral range is determined.

The report starts with a brief theoretical description of the devices and configurations used. Then, the setup is described. Afterwards, the results are presented and discussed.

1 Theory

In this section, the necessary theoretical background of the experiment is provided. This includes the optical elements that were used, a laser diode, an optical grating, a Fabry-Perot interferometer and a photodiode. Additionally, two configurations to achieve optical feedback are described. If not stated otherwise, the content refers to the script [2].

1.1 Diode laser, optical grating, Fabry-Perot Interferometer and photodiode

A diode laser is made of a gain medium, which consists of a N-doped and P-doped semiconductor junction, set under current, provided by an energy source. The input current achieves population inversion by creating electron-hole pairs in the conduction and valence band. The stronger the input current, the more electron-hole pairs are created. If the electrons return to their ground state, they emit energy in the form of photons, by spontaneous or stimulated emission. The frequencies of the photons are spread over a range, determined by the band gap and the current. Onto two opposite sides of the diode a reflective coating is applied to create an optical cavity. Only certain modes are supported by the cavity, leading to increase intensity due to stimulated emission in these modes. The modes can be described by

$$L = n \cdot \lambda_n / 2, \tag{1}$$

where L is the cavity length, n an integer number, and λ_n the corresponding wavelength.

 A diffraction grating uses constructive interference to diffract an incident beam into multiple diffraction orders. The different diffraction orders m obey the equation

$$m \cdot \lambda = d \cdot (\sin \alpha + \sin \beta), \quad (2)$$

where λ is the wavelength of the incident beam and β is the output angle of the m^{th} order.  This formula can be derived using geometric considerations and is discussed thoroughly in many textbooks [3].

A Fabry-Perot interferometer (FPI) is an optical cavity, made of two parallel partly-reflecting mirrors. The mirrors force the light waves inside to have nodes on both ends, leading to discrete modes to be allowed. Thus, it has a wavelength selecting effect. An important property is the free spectral range (FSR), which is the frequency difference between two modes in the cavity, calculated by

$$\Delta\nu_{\text{FSR}} = \frac{c}{2L}, \quad (3)$$

where c is the speed of light in the medium of the cavity.

A photodiode is a device which measures the intensity of light. It essentially is a laser diode operated in reverse. Instead of using electrical current to create photons, an incident photon creates an electron-hole pair in the junction, leading to a small current that can be measured. This current is then proportional to the intensity of the incident light.

1.2 Optical feedback with Littrow and Littman configuration

The Littrow and Littman configuration are two experimental setups created to tune and stabilize a laser's wavelength. In both setups the laser beam is directed onto a diffraction grating such that a certain diffraction order of the laser beam backpropagates into the laser diode. This leads to stimulated emission in the laser medium, amplifying the intensity of this mode in the output laser light. A schematic picture of the Littrow configuration is shown in Figure 1. The collimated beam of the laser diode is directed onto the diffraction grating so that the -1^{st} order is directed back into the laser diode to amplify the intensity of this mode by stimulated emission.

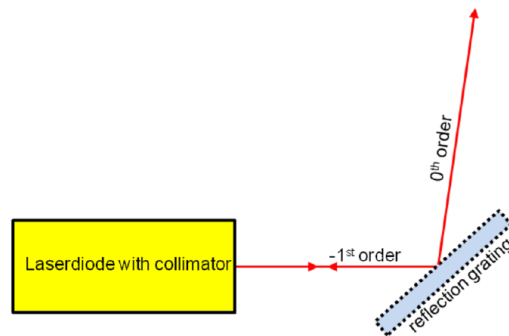


Figure 1: A schematic picture of optical feedback with Littrow configuration, including a laser diode and a reflection grating, which reflects the -1^{st} order back into the laser diode. Figure taken from [2].

In the Littman configuration the -1^{st} order is not redirected immediately into the diode but instead is reflected by mirror, as shown in Figure 2. From the mirror, this order is reflected back onto the grating, and afterwards to the diode. By slightly adjusting the distance to the mirror, the amplified mode of the laser can be switched.

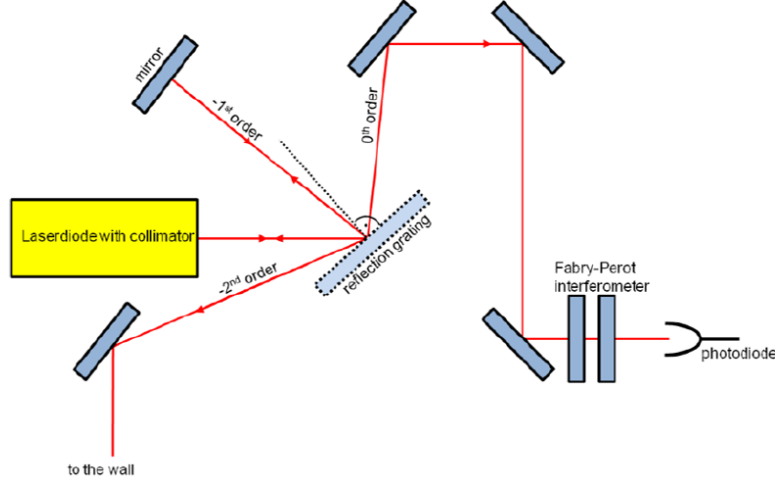


Figure 2: Picture of the Littman configuration. The laser beam is directed onto a grating, which reflects the -1^{st} order onto a mirror, which reflects it back onto the grating and the diode. The 0^{th} order is in our setup redirected by three mirrors through a Fabry-Perot interferometer and entering a photodiode. Figure taken from [2].

2 Setup

In the first part of the experiment, the diode laser (Thorlabs HL6756MG) is characterized without optical feedback. To measure the output power, a powermeter is located in front of the laser. The power is measured for different input currents, which are varied from 0 mA up to 35 mA.

Secondly, the Littrow configuration is realized. A picture of the setup is shown in Figure 1. In order to align the grating, the current is set slightly below the measured threshold current. The 0^{th} order reflection is captured on a sheet of paper. The screws and the angle of the grating are aligned, until the beam flashes, which means the intensity is increasing rapidly. If this occurs, the feedback is achieved. Then, a picture is taken to measure the angles, and the characteristic curve is measured by ~~locating the powermeter~~ in the 0^{th} order reflection and varying the current as before.

Then, the Littman configuration is achieved, as shown in Figure 2. The paths between the diode laser and the grating, as well as between the grating and the mirror, should not exceed 10 cm. The 2^{nd} order is directed onto a sheet of paper, located 210(2) cm far from the mirror. The uncertainty is estimated due to possible distortions in the measuring tape. Using the beam on the screen, the screws on the mirror are adjusted so that clearly distinguishable modes can be seen on the sheet of paper. When the modes are clearly separated and one mode is dominant, the Littman configuration is achieved. Then, the characteristic curve is recorded in the same

manner as before.

Next the distances between the different modes on the sheet of paper are measured. To estimate the uncertainty and reduce systematic errors in the measurements, both students do the measurements independently of each other.

Finally, we determine the FSR of a Fabry-Perot interferometer using the full setup shown in Figure 2. While the 2nd order diffraction is still pointing on the sheet of paper, the 0th order is redirected by two mirrors and enters a FPI, of a length less than 1 mm. The FPI is coupled to a piezo which varies the length of the cavity. The beam passes an attenuator, then it enters a photodiode. The photodiode and the piezo are connected to an oscilloscope. The voltage is recorded for different modes, which can be switched by adjusting the mirror back-reflecting the -1th order. On the sheet of paper, the switch of modes can be recognized.

3 Results

In this section, the results are presented. The section is split into different subsections corresponding to the different experimental setups.

3.1 Diode laser without optical feedback

In the first part of the experiment, the characteristic curve of the laser diode used was measured without any optical feedback. A powermeter is located in front of the laser diode, recording the output power P_{out} for different input currents I_{in} of the diode. The current was varied in a range from 0 mA to 35 mA.

The characteristic curve, which is the output power of the laser ~~dependent~~ for varying input current, is shown in Figure 3. In order to determine the threshold current where the lasing process starts, two linear fits are performed. One describes the behavior of the output power in the current region below the threshold, the other one above the threshold. The data points close to the threshold are neglected as neither spontaneous nor stimulated emission is the dominant emission. The data points are color-coded such that blue data points correspond to the fit below the threshold and red ones above it. White data points were not used for the fits. Both regimes are very well described by a linear fit. However, in the lasing region, for high currents above approximately 29 mA, strong deviations from the linear behaviour occur, which is the reason why these points were also neglected for the fit. The deviation is possibly the result of saturation effects of the laser diode. The lasing threshold current is determined as the intersection of the two linear functions as

$$I_{\text{th}} = 20.28(5) \text{ mA}. \quad (4)$$

Gaussian error propagation is used to estimate the uncertainty for the threshold from the statistical uncertainties calculated by the fitting routine. In Figure 3, the errors of individual datapoints cannot be made out as they are too small. They are estimated from the fluctuations of the display of the power or current measurement. The error of the current is estimated 0.2 mA, the power's uncertainty as 1 μ W.

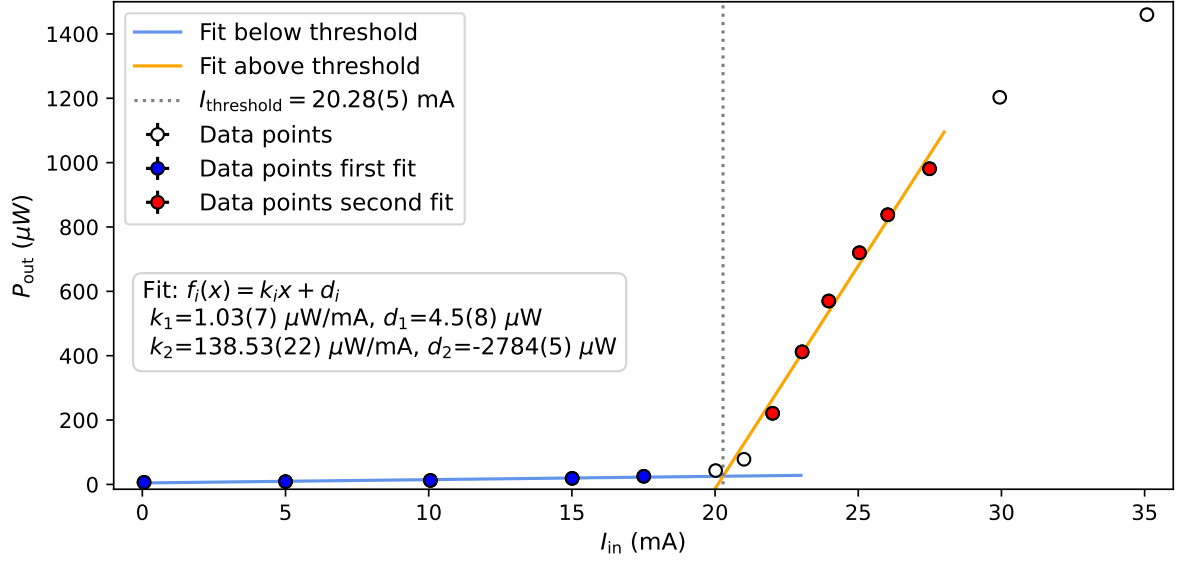


Figure 3: The characteristic curve of the laser diode without optical feedback is plotted. Two linear fits are created, shown in blue (by the blue data points) and in orange (by the red data points). The x -coordinate of the intersection of both fit-functions is shown as a vertical line.

3.2 Diode laser in Littrow configuration

In this part of the experiment, optical feedback is achieved in the Littrow configuration. The characteristic curve is measured in a similar manner to subsection 3.1. The characteristic curve is shown in Figure 4. The procedure of determining the threshold current is equal to the case without feedback. The threshold current is slightly smaller than without feedback:

$$I_{th,Littrow} = 19.17(8) \text{ mA}. \quad (5)$$

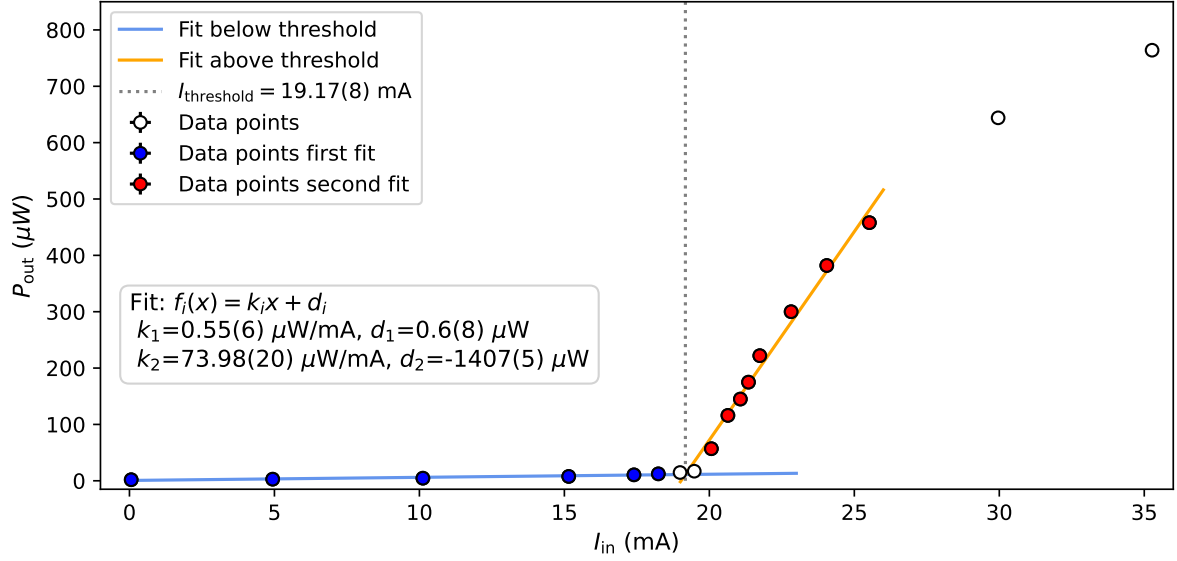


Figure 4: The characteristic curve of the laser diode in Littrow configuration is plotted. The two fits are created as in Figure 3.

The next step is to determine the wavelength λ using the grating equation. The blaze of the grating is $d = 1200 \text{ mm}^{-1}$. Additionally, the input angle of the laser to the grating, which is also the angle of the outgoing -1^{st} reflection order, has to be determined in the experimental setup. It is measured by taking a picture of the setup and measuring all angles in this picture, as shown in Figure 5.

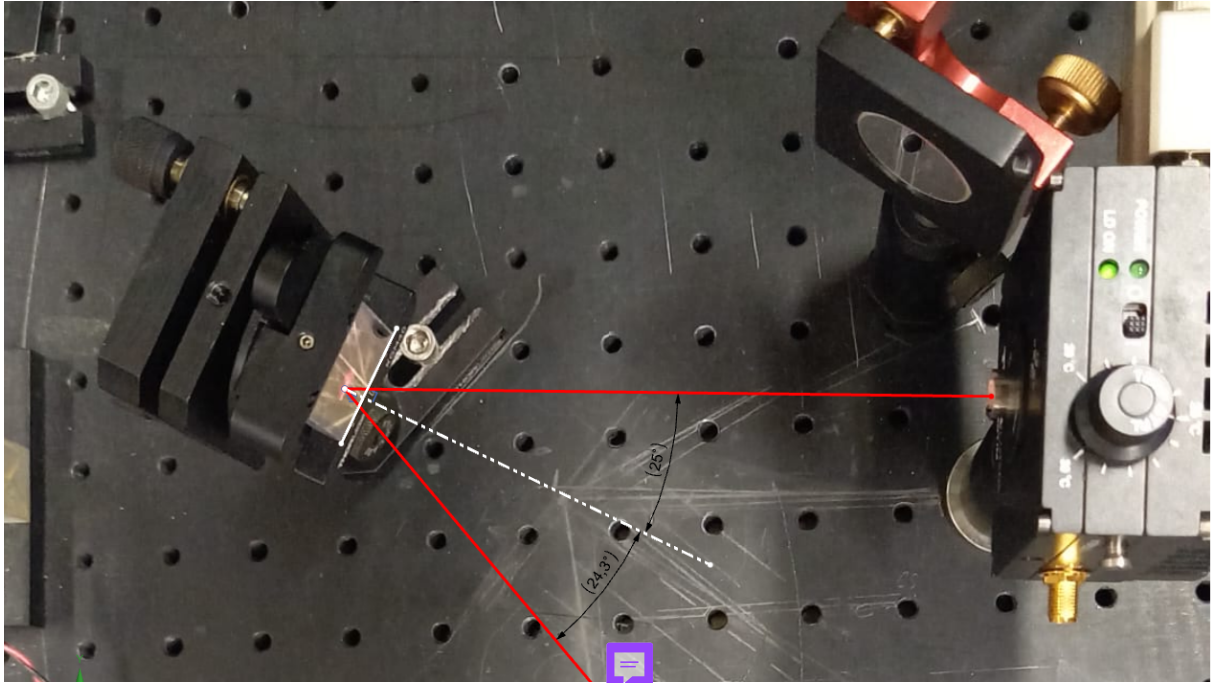


Figure 5: Picture of the setup of the diode laser with feedback in Littrow configuration to determine the relevant angles. A line normal to the grating is drawn (white), as well as the direction of the laser beam (red). The incident angle $\alpha = 25(3)^{\circ}$ as well as outgoing angle $\beta = 24(3)$ is determined using a computer aided modeling (CAD) tool.

In the Littrow configuration, the input angle of the beam and the output angle of the -1st order is the same, which simplifies the grating equation Equation (2) to

$$\lambda = 2d \sin \alpha. \quad (6)$$

Inserting d and the measured angle $\alpha = 25(3)^\circ$ leads to a wavelength of

$$\lambda_{\text{Littrow}} = 704(79) \text{ nm}, \quad (7)$$

which is in accordance with the wavelength $\lambda = 670 \text{ nm}$ given by [1]. The uncertainty in α is estimated as 3° , as many factors can cause errors in the angle measurement. For one the normals on diode laser and grating cannot be drawn very precisely. Additionally, we have to account for projection effect caused by the finite distance of the phone, as well as possible tilting with regards to the optical table. 

3.3 Diode laser in Littman configuration

In the Littman configuration, the characteristic curve is measured again, with the same procedure as in the two tasks before. The characteristic curve is plotted in Figure 6, the intersection of the two fits yields a threshold current of

$$I_{\text{th,Littman}} = 21.01(11) \text{ mA}, \quad (8)$$

which is slightly bigger than in the Littrow configuration.

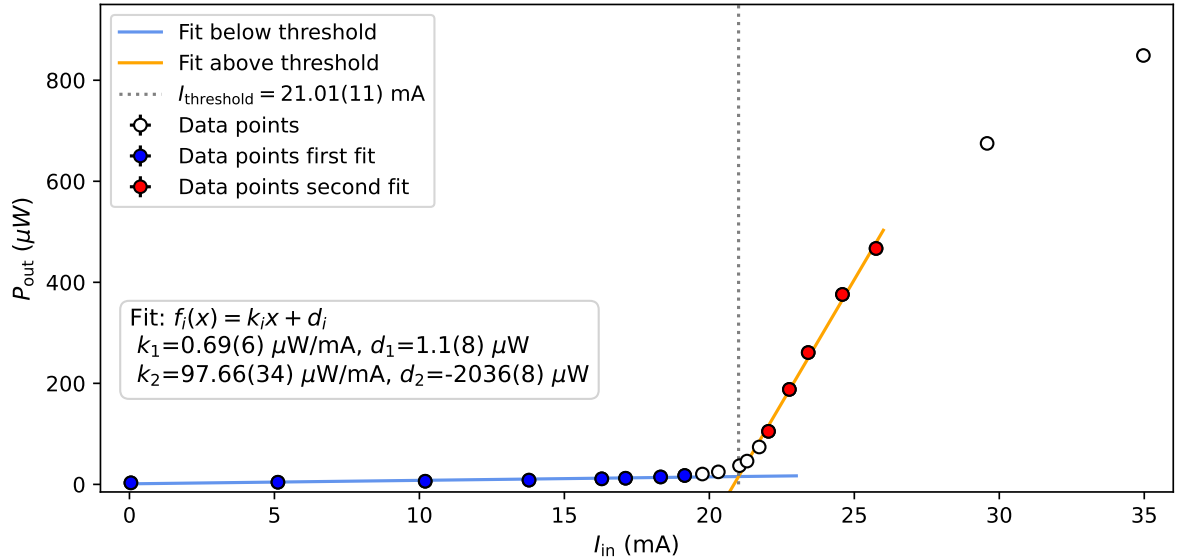


Figure 6: The characteristic curve of the diode laser in Littman configuration with two linear fits for the two regimes. The threshold current is determined as the intersection and is $21.01(11) \text{ mA}$.

The wavelength is determined again with Equation (2), the measured input angle $\alpha = 45(3)^\circ$ and the output angle $\beta = 8(3)^\circ$, which were measured from the picture, given in Figure 7. This

yields

$$\lambda_{\text{Littman}} = 672(12) \text{ nm}, \quad (9)$$

which again is consistent with a wavelength of 670 nm. Compared to the Littrow configuration the statistical uncertainty is much smaller.

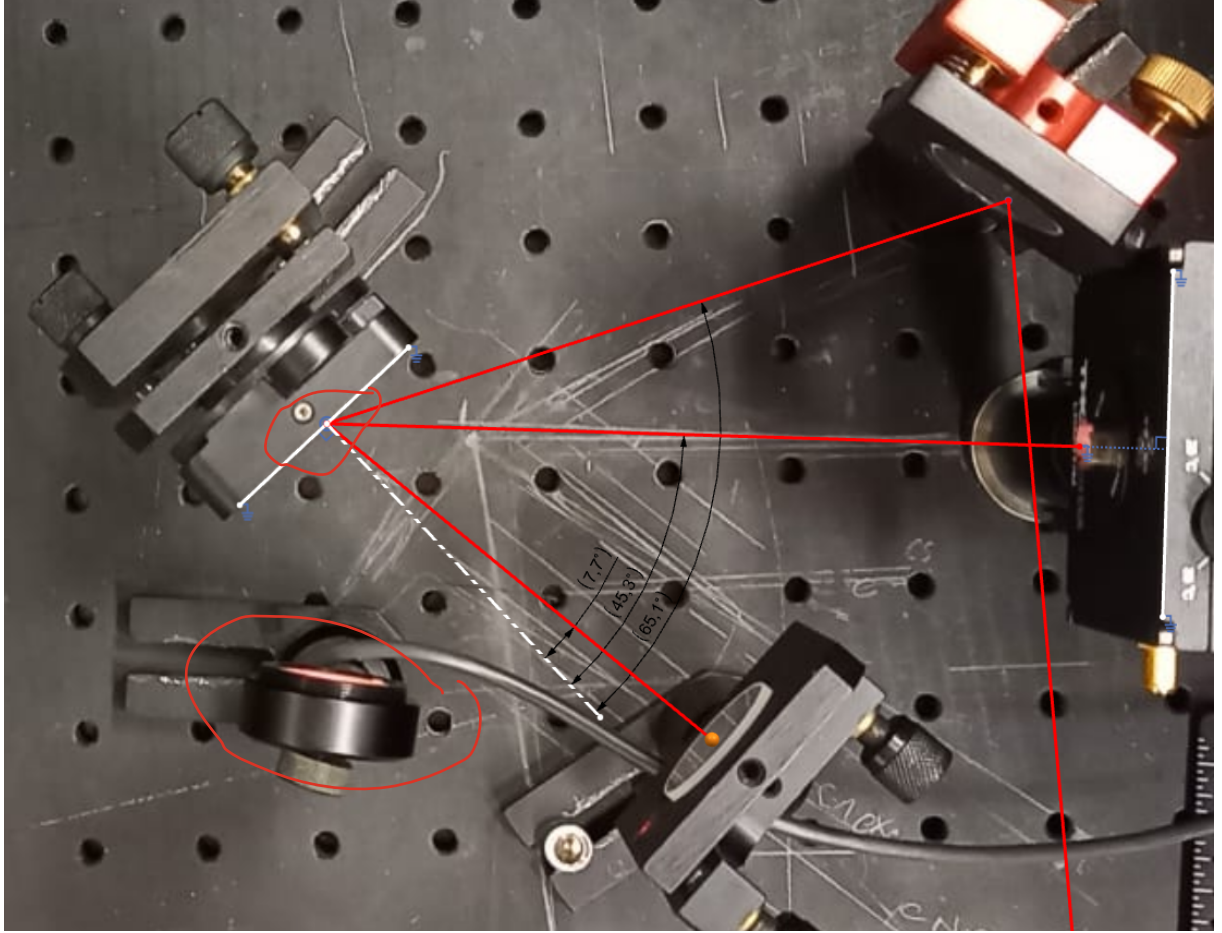


Figure 7: Setup of the Littman Configuration. The incident angle of the laser beam and the diffraction angle of the outgoing -1st order. The angles $\alpha = 45(3)^\circ$ (incoming), $\beta_1 = 8(3)^\circ$ (-1st order) and $\beta_2 = 65(3)^\circ$ (-2nd order) are measured relative to the normal of the grating.

3.4 Wavelength tunability

To investigate the tunability of the laser the -2nd diffraction order was projected onto a screen at a distance of $D = 210(2)$ cm. We notice distinct intensity peaks corresponding to the different laser modes supported by the laser diode cavity. One of the many modes is significantly brighter than the others and we can switch the dominant mode by adjusting the angle of the diffraction grating, effectively tuning the laser.

We mark the observable lasing modes on a paper and measure the distance between them using a ruler. These measurements are performed independently by two students and the mean is taken to reduce systematic deviations. In this way we are able to measure 15 distinct lasing modes. We set the central mode as reference and calculate the cumulative distance to it. We can relate the projected displacement Δx to a change in angle $\Delta\beta$ using the distance from

the grating to the wall D and trigonometric considerations (and assuming angles to be small), leading to

$$\Delta\beta = \arctan\left(\frac{\Delta x}{D}\right). \quad (10)$$

The measured projected displacement as well as calculated angle difference can be seen in Table 1. A change in incident angle $\beta_2 + \Delta\beta$ leads to a change in frequency, given by

$$\Delta\nu = c \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_0} \right), \quad (11)$$

with the frequency $\lambda_0 = 672(12)$ nm and the changed frequency $\lambda_1 = \frac{d}{2}(\sin(\alpha) + \sin(\beta_2 + \Delta\beta))$ calculated from the grating equation. Here $d = 1200$ 1/mm refers to the blaze of the grating. The calculated frequency shift (again relative to the central mode) is also included in Table 1. However we do not include its statistical uncertainty due to its large value (~ 100 GHz). 

Table 1: The measured projected displacement Δx , as well as the corresponding angular change $\Delta\beta$ are shown in this table. Furthermore the induced frequency shift $\Delta\nu$ is included without the statistical uncertainty as its large value would dominate, washing out the small calculated frequency changes between neighboring modes. 

mode	Δx (mm)	$\Delta\beta$ (°)	$\Delta\nu$ (GHz)	mode	Δx (cm)	$\Delta\beta$ (°)	$\Delta\nu$ (GHz)
-7	-11.78(19)	-0.321(6)	-660	1	2.00(7)	0.055(2)	111
-6	-10.15(17)	-0.277(5)	-568	2	3.50(10)	0.095(3)	194
-5	-8.15(16)	-0.222(5)	-456	3	4.75(12)	0.130(4)	263
-4	-6.40(14)	-0.175(4)	-357	4	6.50(14)	0.177(4)	360
-3	-4.90(12)	-0.134(4)	-273	5	8.12(16)	0.222(5)	450
-2	-3.25(10)	-0.089(3)	-181	6	10.12(17)	0.276(5)	560
-1	-1.75(7)	-0.048(2)	-97	7	12.12(19)	0.331(6)	669

Using the values in Table 1 we can also approximate the tunable range of the laser without loosing feedback. For this we subtract the induced frequency difference between the modes furthest away from the central reference mode, which were still visible. We get a tunable range of the laser of 1330(130) GHz. This range can be improved by paying closer attention to aligning the mirrors, as we determined empirically that correct alignment is crucial in this experiment.

3.5 Free spectral range

Lastly we couple the 0th diffraction order via mirrors to a Fabry-Perot interferometer (FPI). Using the known frequency shift when switching between lasing modes we can measure the free spectral range (FSR) of the FPI, a fundamental quantity of the cavity. For this we record timetraces of the driving ramp (to make sure we capture a full piezo sweep of the cavity), as well as the outgoing light captured by a photodiode. We record traces for four different lasing modes and the data can be seen in Figure 8 (all traces can be seen as gray dots).

If the beam is correctly aligned with regards to the confocal cavity we expect Lorentz profiles corresponding to TEM₀₀ modes of the cavity. Looking at the resonance curves in Figure 8

we notice that the peaks are skewed and are better represented by Gaussian profiles. Since we are only interested in the position and width of the peaks, we fit functions of the form $A \exp(-\frac{(x-x_0)^2}{2\sigma^2}) + c$ with the scaling A , center x_0 and width σ as free parameters. The vertical offset c was not included in the fit but instead calculated using the mean calculated from background noise.

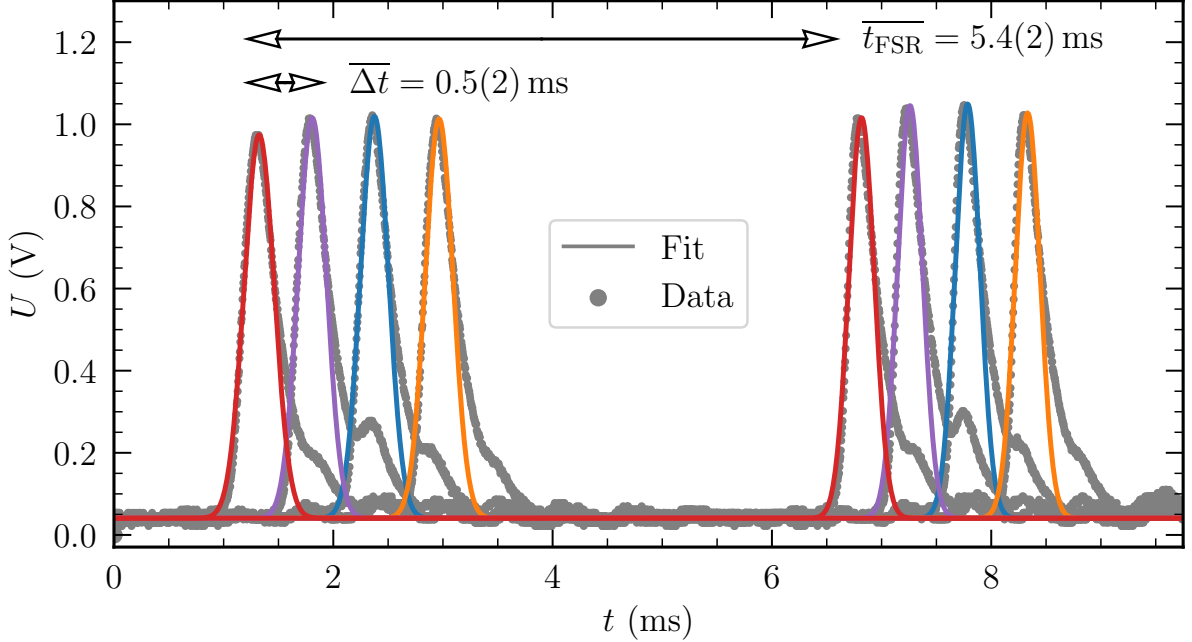


Figure 8: Timetraces of four different lasing modes coupled to an FPI are recorded using an oscilloscope and are plotted as gray dots. Gaussian functions are fitted to the resonance peaks and can be seen as solid lines (different timetraces are colorcoded). From the fitted peaks we can calculate the time-domain free spectral range, as well as the timeshift induced by switching modes.

For each timetrace we calculate the time-domain free spectral range t_{FSR} by subtracting the two peak positions (setting the uncertainty of a peak to the full width at half maximum of the Gaussian fit). Since t_{FSR} is expected to remain constant for all lasing modes we take the mean to reduce the statistical uncertainty, leading to $\overline{t_{\text{FSR}}} = 5.4(2)$ ms. We further calculate the difference between neighboring peaks Δt and again average over all measured values since the frequency shift w.r.t. $\Delta\beta$ also behaves linearly (not exactly but can be assumed for small angles), leading to $\overline{\Delta t} = 0.5(2)$ ms.

Lastly we calculate the mean frequency shift between neighboring lasing modes using the values in Table 1 $\overline{\Delta\nu} = 95(9)$ GHz. Now we can calculate the FSR of the FPI by converting the time-domain FSR, rescaled by the time-to-frequency conversion factor, summarized in the following equation:

$$\nu_{\text{FSR}} = \frac{\Delta\nu}{\Delta t} t_{\text{FSR}}. \quad (12)$$

In the end we calculate $\overline{\nu_{\text{FSR}}} = 980(210)$ GHz, which is in accordance with the given range of 500 – 1000 GHz in the datasheet of the FPI [2]. Instead of averaging all time-displacements due to mode jumping we could have connected a certain mode jump (from Figure 8) to a specific

frequency shift (from Table 1), which from a theoretical standpoint makes more sense since the underlying physics is nonlinear rendering all mode-jumps distinct. However we deemed a linear approximation for our experimental setup beneficial (this can be argued by noticing that all angular changes considered in the experiments are small), as it allows us to work with mean values, reducing the statistical uncertainty.

4 Conclusion

In this report we investigated the effect of optical feedback when using a laser diode with three different setups. We start by characterizing the laser diode without any feedback and extract the lasing threshold current $I_{\text{th}} = 20.28(5) \text{ mA}$. We proceed by using a diffraction grating to couple the -1^{st} diffraction order back into the laser diode to achieve optical feedback in the Littrow configuration. In this setup we extract an threshold current of $I_{\text{th,Littrow}} = 19.17(8) \text{ mA}$ and a laser wavelength of $\lambda_{\text{Littrow}} = 704(79) \text{ nm}$ using the grating equation and angle measurements. Lastly we use a mirror to reflect the -1^{st} diffraction order back onto the grating to achieve feedback in the Littman configuration. The threshold current $I_{\text{th,Littman}} = 21.01(11) \text{ mA}$. The wavelength in the Littman configuration is $\lambda_{\text{Littman}} = 672(12) \text{ nm}$. Both wavelengths of Littrow and Littman setups are consistent with the wavelength given in [1], however the Littman configuration is more accurate with the uncertainty smaller by a factor of 7. To further reduce the statistical uncertainty one may want to conduct more precise angle measurements in followup experiments.

Next we investigate the wavelength tunability of the laser in the Littman configuration by hopping between internal lasing modes of the diode while retaining optical feedback. For this we project the -2^{nd} diffraction order onto a screen and mark the distinct lasing modes which we can access by rotating the grating. We can relate the measured projected distance to an angular shift, which in turn is caused by a shift in frequency according to the grating equation. We measure a mean mode spacing of $\overline{\Delta\nu} = 95(9) \text{ GHz}$ and a tunable range without losing optical feedback of $1330(130) \text{ GHz}$. More careful adjustment and measurement of angles can reduce the statistical uncertainty, as well as increase the tunable range of the laser.

Lastly we couple the 0^{th} diffraction order to a Fabry-Perot interferometer driven by a piezo element and record the transmitted light with a photodiode. We record timetraces for four different lasing modes and extract peak positions using Gaussian fits. From here we calculate the free spectral range in the time domain, as well as the time shift when switching modes. Using these measurements we can convert the time-domain FSR to frequencies, leading us to a value of $\nu_{\text{FSR}} = 980(210) \text{ GHz}$. This is in accordance with the possible range of the FSR of $500 - 1000 \text{ GHz}$ given in [2].

References

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Erklärung

Hiermit versichern wir, dass der vorliegende Bericht selbständig verfasst wurde und alle notwendigen Quellen und Referenzen angegeben sind.



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Alexander Helbok

..... January 8, 2025
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Date



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Max Koppelstätter

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Date